BRIDGES IN MATHEMATICS • GRADE 3 • UNIT 5

Multiplication, Division & Area: Open Arrays

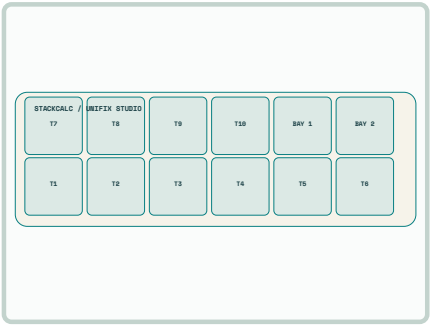
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.3.MD.C.7

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Unifix Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Build a 6×7 array in the Counting & Array Studio. Split into a 6×5 section and a 6×2 section.

Sketch Model on Paper:
Draw an open array partitioned into 6×5 and 6×2 rectangles.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	6	X: 6	Dimension 6 into X
2.	ENTER	Y: 6 | X: 6	Stack lift
3.	5	Y: 6 | X: 5	Dimension 5 into X
4.	×	X: 30	1st partial area: 30

EXPECTED LCD X: 42 sq units

1 **Split 4 × 8 into (4×5) + (4×3).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Split 7 × 6 into (7×5) + (7×1).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Find total area of 8 rows of 9 squares.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Area is 36 sq units, width is 4. Find length.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 5

Multiplication, Division & Area: Open Arrays • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Counting & Array Studio (COUNT-01)

CAD Blueprint: unifix_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/unifix_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Charcoal Array Tray & Teal Counting Cubes (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Split 4 × 8 into (4×5) + (4×3).** Standard: CCSS.MATH.3.MD.C.7	32	[4] [ENTER] [5] [×] [4] [ENTER] [3] [×
2	**Split 7 × 6 into (7×5) + (7×1).** Standard: CCSS.MATH.3.MD.C.7	42	[7] [ENTER] [5] [×] [7] [ENTER] [1] [×
3	**Find total area of 8 rows of 9 squares.** Standard: CCSS.MATH.3.MD.C.7	72 sq units	[8] [ENTER] [9] [×]
4	**Area is 36 sq units, width is 4. Find length** Standard: CCSS.MATH.3.MD.C.7	9 units	[3][6] [ENTER] [4] [÷]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for adding dimensions instead of multiplying to find area.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.